



Land Management Training Center

Introduction to Digital Cadastre

What is Cadastre?

- “A cadastre is normally a parcel based, and up to date land information system containing a record of interests in land (e.g. rights, restrictions and responsibilities)”.
- It usually includes a geometric description of land parcels linked to other records describing the nature of the interests, the ownership or control of those interests, and often the value of the parcel . (FIG 1995)
- It gives an answer to the each property Where and How much.

What is Cadastre?

The cadastres is an information system consisting of two parts;

- *A series of maps* showing the size and allocation of all land parcels together with
- *Text records* that describe the attributes i.e. characteristics of the land.



Cadastral

- *Cadastral surveys* document the boundaries of land ownership with the help of documents, diagrams, sketches, plans, charts, and maps.
- They were originally used to ensure reliable facts for land valuation and taxation.
- An example from early England is the Domesday Book.
- Napoleon established a comprehensive cadastral system for France that is regarded as the starting of most modern versions

Types of Cadastre

Fiscal Cadastre:

A cadastre, which is mainly established for the purpose of revenue and tax collection

Legal Cadastre:

A cadastre which is established to guarantee the public right and restriction over the land parcel. It explains the legal status of the

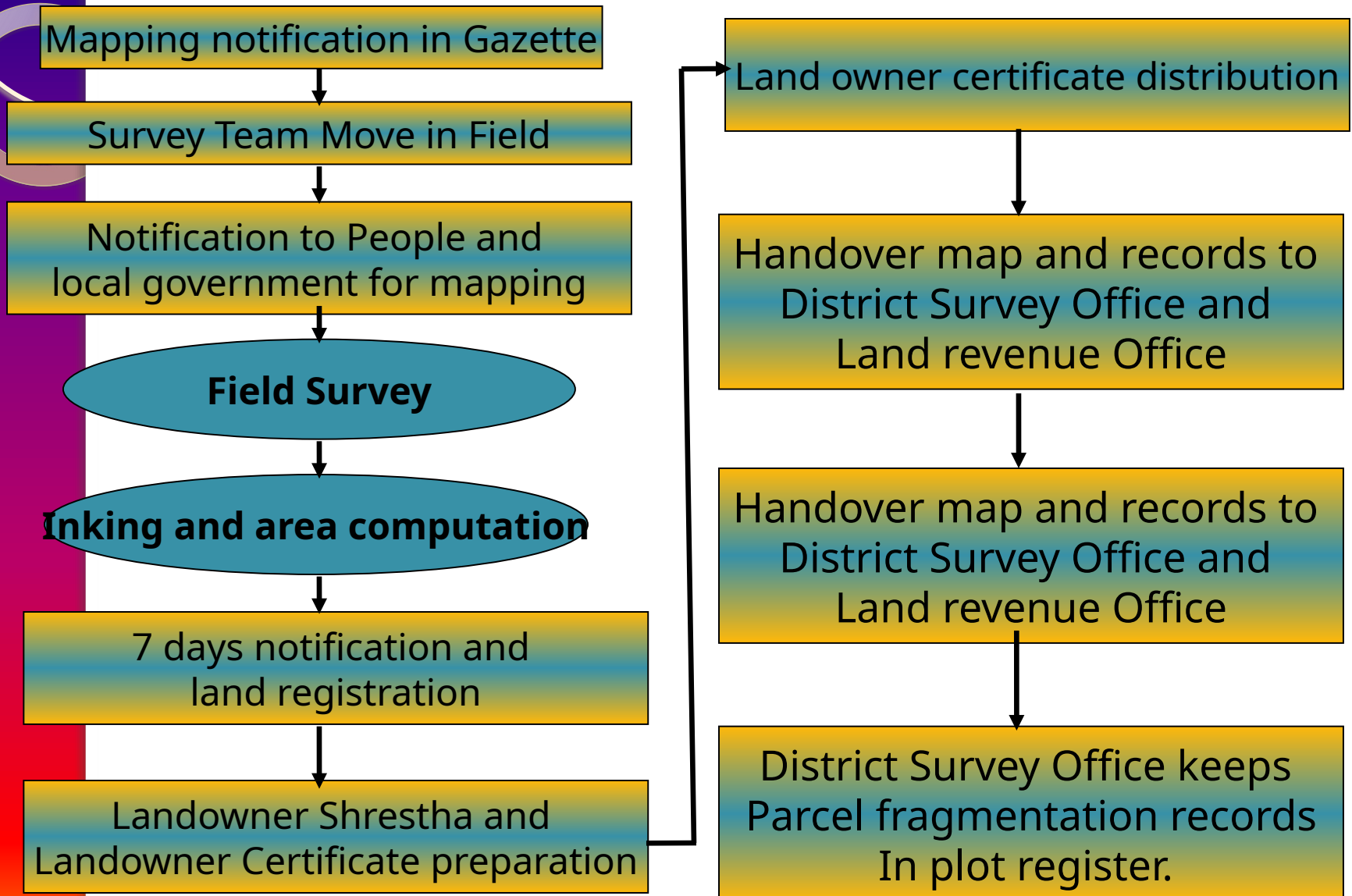
Cadastre

Multipurpose Cadastre:

A cadastre which is established not only to show fiscal and legal status of the land, but also shows other geo data like topographic and land use data. It can be used for different infrastructure development, planning and construction activities, also known as

Land Information

Methodology of Analog Cadastral System



Initial Field Book (Prarambhic Terij)

श्री ५ को सरकार
भूमि सुधार तथा व्यवस्था मन्त्रालय
नापी विभाग
६ नं. नापी गोश्वारा
प्रारम्भिक तैरिज

जिला : जयपुर
गा. वि. स. दुबेयू. गाराणा
वाट नं. ८

सि. नं.	जमात धनीको नाम धर वतन	वर्तन भाएको कित्ता	वर्तन बाँकी कित्ता	वैकित्यत
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२	-	साविक सिलापाडला ५४ १२८५, १२८० १२८९, १२८२	२	अधुरो
३	-	साविक सिलापाडला ५४ ६२८, ६३०	३	अधुरो
४	-	साविक सिलापाडला ५४ वि. नं. ६३९	४	
५	-	-	५	अधुरो
६	-	-	६	अधुरो
७	-	साविक सिलापाडला ५४ वि. नं. ६२६	७	अधुरो
८	-	साविक सिलापाडला ५२९, ५२३	८	अधुरो
९	-	साविक सिलापाडला ५२९ वि. नं. ६१८, २१५, २१०	९	
१०	-	साविक सिलापाडला ५४ वि. नं. १६८३	१०	
११	-	साविक सिलापाडला ५४ वि. नं. १६८४	११	
१२	मुसा, गार ५१ नं. ५४ ८ कपिल / कोजको गार	साविक सिलापाडला ५४ वि. नं. १०३१ १०३४	१२	
१३	५१ - ५४ गार ५१ - ५४ वि. नं. १०३१, १०३४ वर्तन २२			

५१ - ५४ गार
५१ - ५४ वि. नं. १०३१, १०३४
वर्तन २२

[illegible]

Analog system of Nepalese cadastre.....

8-2-99-2-02

नापी क्षेत्रीय किताब

नापी गोश्वारा नं. ६२

नापी टोली नं. १

संक्षेपतः

जिल्हा

महा/उपमहा/नगरपालिका

वडा

माला पत्र नं. १२२-११२२ दि. नं. १ ईति

नापी पर्वे संपादन

निरीक्षण

टोली प्रमुख

काम शुद्ध मिति

काम समाप्त मिति

2007 6 19

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माला पत्र नं. १२२-११२२ दि. नं. १ ईति

नापी पर्वे संपादन

निरीक्षण

टोली प्रमुख

काम शुद्ध मिति

काम समाप्त मिति

2007 6 19

Field Book

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 सिद्ध है :-
 सिद्ध है :-

श्री ५ को सरकार
भूमिसुधार तथा व्यावस्था मन्त्रालय
नापी विभाग

किता नापी महाशाखा

६. नं. नापी गोश्वारा, काठमाडौं

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9. 謝文雄：《論我國行政程序法之制定》，收於《行政程序法之制定與實施》，頁1-10。

발행처

(१) जग्गाको व्यवस्थापन गर्दा रैथानेको श्रेष्ठता र रैथाने सम्बन्धित अन्य विवरण :-

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संस्कृत - ५० अंश

[illegible]

सूचना - बाउसाही उपनका बाउक अन्य क्षेत्रको लागि उपयुक्त भुविछा १० कि.मी. भित्राभन्को लाई मानिनेछ ।

समर्थन :- कृषि उत्पादन को प्रोत्साहित करने के लिए :-

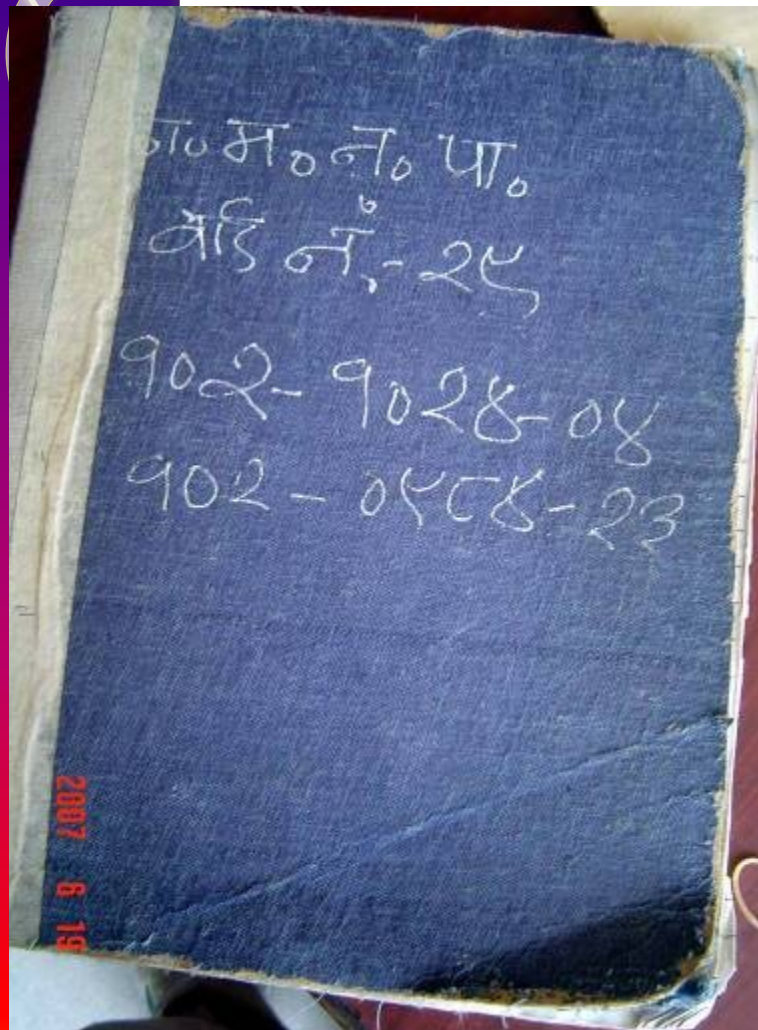
क. 'सांस्कृतिक तथा कलाशास्त्र' प्रमाणों किन्हीं उद्योग, व्यापार, शिक्षा, स्वास्थ्य, सम्पत्ति व मनोरंजन आदि व्यवसाय सम्बन्धित हूँ तथा मैंने किसीका कलाशास्त्र लेखाई नज़दीक।

भाषा विरीलक

नापी टोली प्रमुख

बाली प्रमाण

Analog system of Nepalese cadastre.....

[illegible]

Plot Register

Nepalese context of cadastre.....



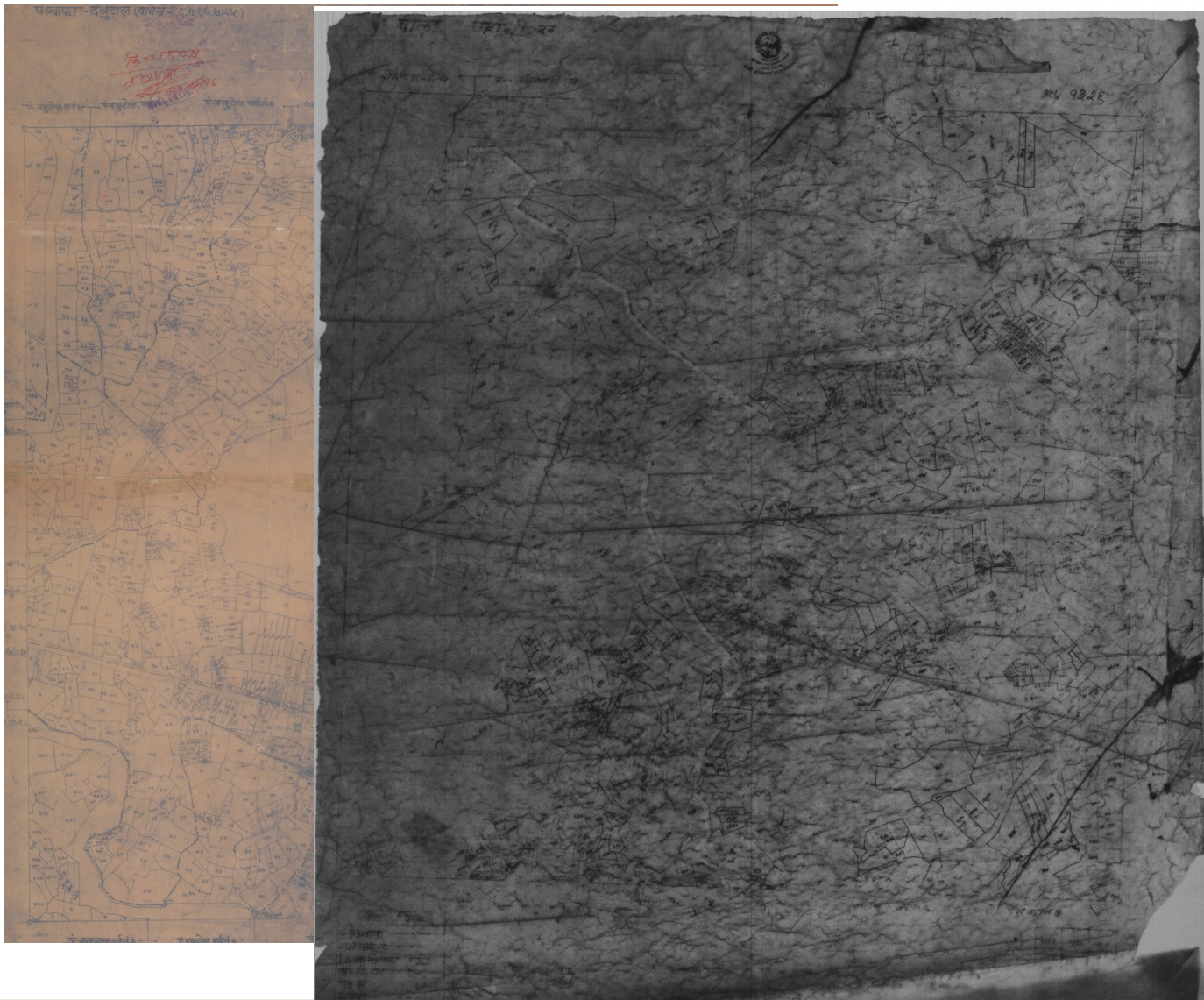
Map Cabinet

Current Situation of cadastral survey

Maps



Cadastral map of Banepa Bazar in 1:2500 Scale



Plot register and other documents





What is international Trend

- Digital land recording system
- Private consultancies are involved
- Fixed boundary system
- Web mapping
- Transaction and sharing through Internet based system
- Fast service delivery
- Digital environment for map updating
- Cadastral Information System (CIS)



Digital Cadastre

- The process of cadastral survey in which each corner points of any land parcel are first identified , adjudicated demarcated and measured to compute the coordinate of each corner point is called Digital Cadastre.
- This is more accurate and used to those areas where land value is very high.
- The data obtained in this process are processed by using computer and then plotted to any required scale.



Why digital cadastre?

- For accurate & reliable information
- For more secure & easily accessible data.
- For the development of multi purpose cadastre system.
- For reducing land disputes
- For addressing Cadastral Vision 2014
- To fulfill users need & satisfaction



Why digital cadastre?

- For Effective & efficient land management & land administration.
- For effective service delivery in digital environment
- To develop Parcel Based Land Information System
- For Good Governance

Why Digital Cadastre ?



Fulfill demand of public



Fast Service Delivery

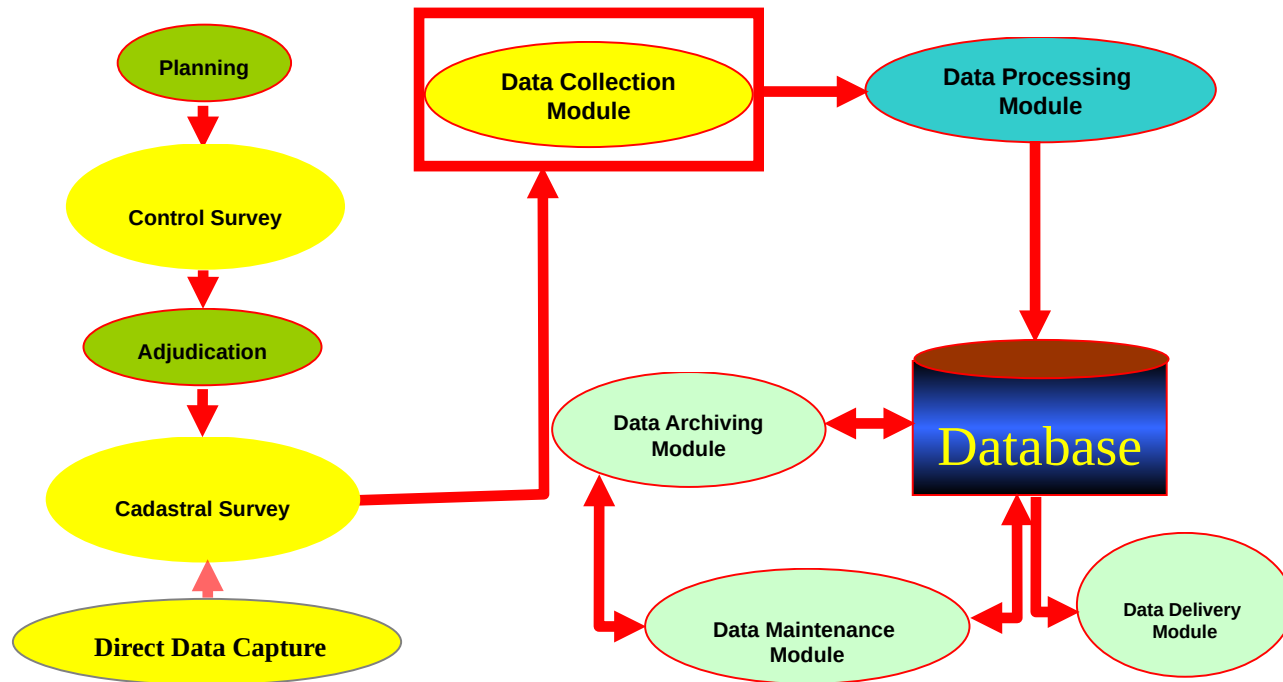


Development of technology



To establish parcel based land
information system

Frame work of Digital Cadastral system





Major Steps of Digital Cadastral survey

- Planning (equipments , control points...)
- Public notice for field surveying
- Control point densification
- Systematic field Adjudication
- Demarcation
- Sketching
- Field Survey (using Total station, GPS)
- Data downloading
- Plotting and database designing
- Mapping



Major Steps of Digital Cadastral survey

- Map verification
- Land registration
- Decision on land disputes
- Land owner certificate distribution
- Hand over the documents to land revenue office and survey office
- Digital service and database updating

Adjudication



Field Observation



Field Sketch



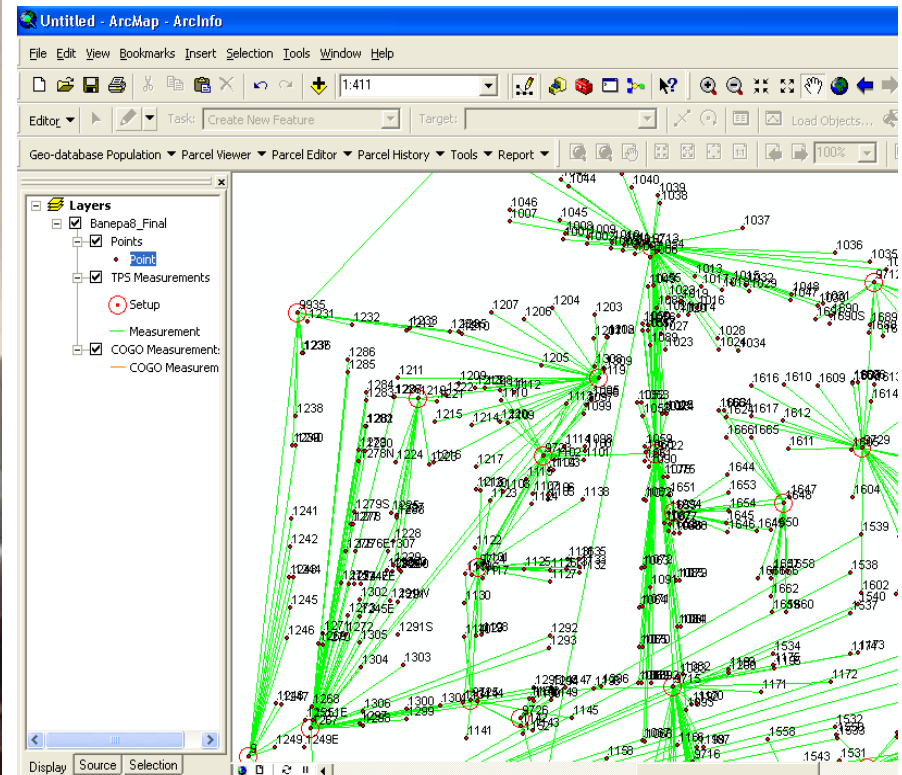

 नेपाल सरकार
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 नापी विभाग
 कित्ता नापी महाशाखा
नापी कार्यालय, बनेपा
 डिजिटल कित्ता नापीको फिल्ड स्केच

छुलिस्वेल
 गारपालिका : बनेपा
 वडा नं. ५
 स्थान/टोल : बनेपा
 मिति : २०७३/०८/२५

फिल्ड स्केच
 १. विष्णु लाल
 २. विष्णु लाल
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 १००. विष्णु लाल

स्केच तयार गर्ने : मोती बादी
 नापी गर्ने : अनिल भट्ट
 चेक गर्ने : अनिल भट्ट
 प्रमाणित गर्ने :

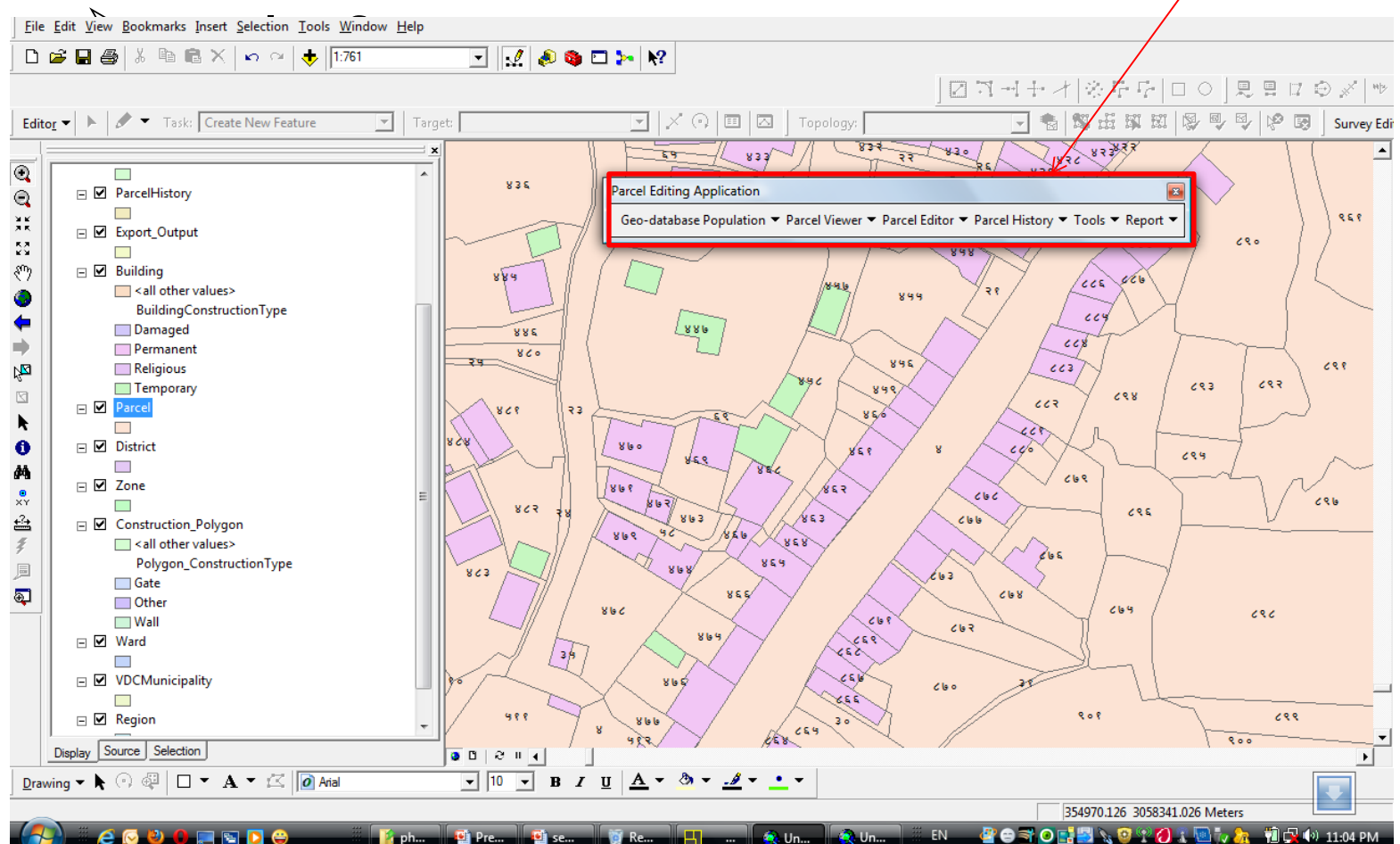
Data download



Database and Mapping

➤ Use of parcel editor

Parcel Editor Menu



Database & Cadastral Map

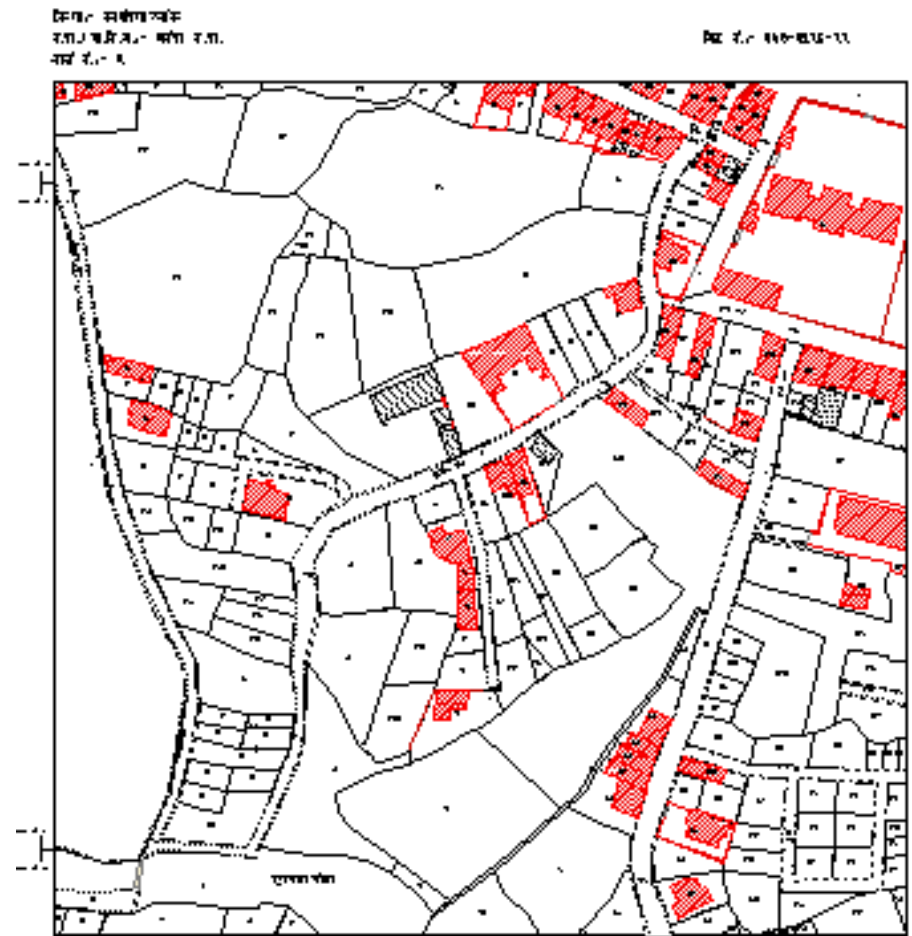
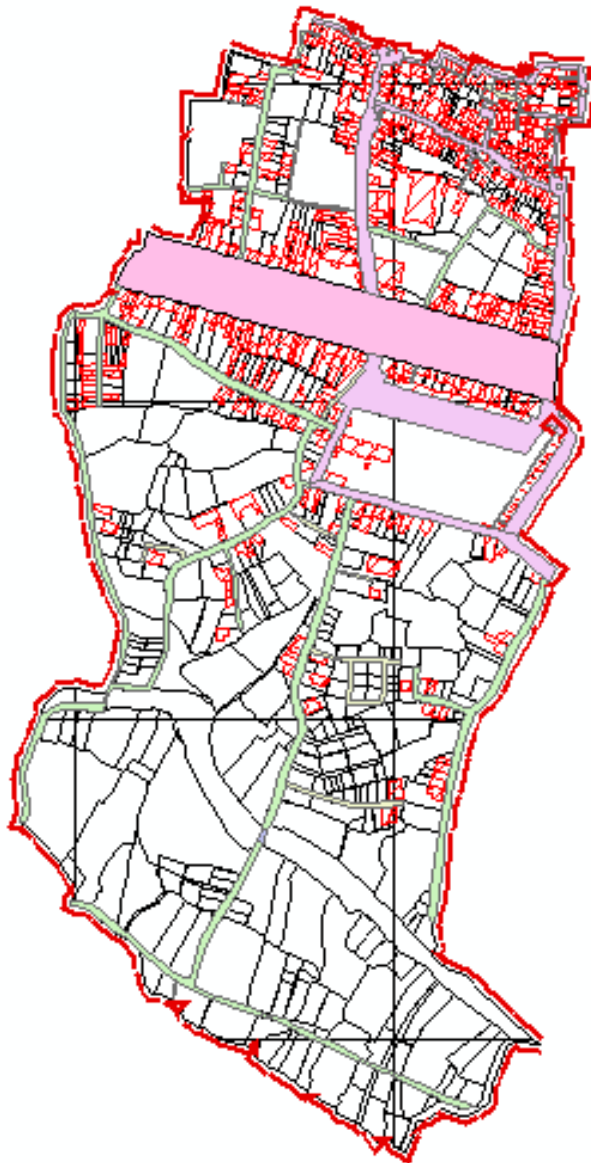


Table 1: A table with 2 columns and 10 rows, likely representing land parcels or survey data. The table is titled 'Table 1' and contains numerical data.

Parcel No.	Area (sq. m)
1	1200
2	1500
3	1800
4	2100
5	2400
6	2700
7	3000
8	3300
9	3600
10	3900

Scale bar: 0 to 100 meters

Table 2: A table with 2 columns and 10 rows, likely representing land parcels or survey data. The table is titled 'Table 2' and contains numerical data.

Parcel No.	Area (sq. m)
1	1200
2	1500
3	1800
4	2100
5	2400
6	2700
7	3000
8	3300
9	3600
10	3900

Legend: A legend explaining the symbols used in the map, including a scale bar and a north arrow.



Ministry of Land Resources, Government of India

Attribute information of selected parcel

The screenshot displays the ArcMap interface with a map of parcels. The 'Layers' panel on the left shows the following layers: ParcelLine, Building, Gridsheet, Parcel, and ParcelHistory. The 'Identify' window is open, showing the attribute information for the selected parcel (922). The 'Identify from' dropdown is set to 'Parcel'. The 'Location' field shows the coordinates 354,244.452 3,057,652.685 Meters. The 'Field' and 'Value' columns list various attributes for the parcel.

Field	Value
OBJECTID	922
SHAPE	Polygon
अञ्चल	बागमती
जिल्ला_कोड	२४
जिल्ला	काभ्रेपलान्चोक
गाविस_कोड	२४००४
गाविस	बनेपा नगरपालिका
वाई	०६
नक्सा_सिट_नम्बर	१५७१३२४२२
जमिनको_कित्ता_नम्बर	१५७
कित्ता_संकेत_नम्बर	१५७१३२४२२०००१५७२४००४०६००००
जग्गाधनी_संकेत_नम्बर	०२१७
किसिम	व्यवसायिक तथा बसोबास
विरह	घर
जग्गाको_क्षेत्रीय_किसिम	शहरी
किसिम_मान	२६
प्रमाण_संकेत	साविक बनेपा ५६ कि.नं. ३५५ क्षेत्र, ०-४-
कैफियत	<null>
दर्ता_भए_नभएको	yes
पूर्व_कित्ता_नम्बर	१
पश्चिम_कित्ता_नम्बर	१५०
उत्तर_कित्ता_नम्बर	१४९
दक्षिण_कित्ता_नम्बर	१५२
नापी_भएको_मिति	२०६२/११/१६
रोपनी	0-4-2-3.47
बिघा	0,0,8
बर्गफिट	1614.60216
SHAPE_Length	57.584846
SHAPE_Area	150.001594
ParcelNoEng	157
AuthenticateID	<null>
दर्ता_गर्नेको_नाम	<null>
नापी_रोली_नं	1
SurveyOfficeName	नापी कार्यालय, बनेपा

Field Book

क्षेत्रीय किताब

नियम ४ के उपनियम (३) प्रग प्रयोजित

जिल्ला:- काठमाडौं

गाउँ विकास समिति/नगरपालिका:- बनेपा नगरपालिका

वडा नं:- ०६

नक्सा फिट नं:- १६७१/०६१७

जमिनको विकास	जग्गाजमाके			मालिक			पाइ (शा.पग.ग मेखी अदि)	विमर्श		अंशफल	जहरी				मिति		प्रमाण प्रकृत	इती गर्नेको नाम र दस्तखत	अक्षर
	नाम, शा. उत्तर, नागरिकता नम्बर तथा गण्डा गज्जेको नाम	ज्येष्ठा	प्रहिले	नाम, शा. उत्तर, नागरिकता नम्बर तथा गण्डा गज्जेको नाम	ज्येष्ठा	प्रहिले		कृषि अंश	अन्यसहित मूल नम्बर		गुं	पश्चिम	उत्तर	दक्षिण	नाथी अक्षर र मिति	मिति			
४१	नाम:- भादकाजी श्रेष्ठ श.पि.प्र:- वनेप- ५ न.प.नं:- १९९ मिति:- २०३९/११/९ जिल्ला:- सर्प गण्डा/पति:- प्रानुसुते गज्जे/प्रमा:- तन्वाम			नाम:- श.पि.प्र:- न.प.नं:- मिति:- जिल्ला:- गण्डा/पति:- गज्जे/प्रमा:-			पञ्चारी		क	२६.७६	३	५	५	५३	२०३९/११/९		वाकिलदहुरोत १/कि नं:६६९, ६६९/ले ५००-९-०-९, ०-९-९-३		
४२	नाम:- वृन्दर क.वी. श.पि.प्र:- वनेप- ३ न.प.नं:- ६४३३५ मिति:- २०४३/०२/०६ जिल्ला:- सर्प गण्डा/पति:- भानु व. क.पी. गज्जे/प्रमा:- दल गज्जे			नाम:- श.पि.प्र:- न.प.नं:- मिति:- जिल्ला:- गण्डा/पति:- गज्जे/प्रमा:-			पञ्चारी		क	१०९.६६	३	५	५	५५	२०३९/११/९		वाकिलदहुरोत १/कि नं:६३९, ६३३/ले ५००-९-९-९, ०-९-३-०		
४३	नाम:- बिपा बस्का श.पि.प्र:- वनेप- ५ न.प.नं:- ३९५४/२३१३ मिति:- २०४०/०४/०४ जिल्ला:- सर्प गण्डा/पति:- निर्मल खड्का/दिवस नाथका गज्जे/प्रमा:- धर्म नाथका			नाम:- श.पि.प्र:- न.प.नं:- मिति:- जिल्ला:- गण्डा/पति:- गज्जे/प्रमा:-			पर		क	७६.६	३	५	६	५५	२०३९/११/९		वाकिलदहुरोत १/कि नं:२०९/ले ५००-९-९-९		
४४	नाम:- कृष्ण बटि बाजुवर श.पि.प्र:- वनेप- ६ न.प.नं:- ३०५४/२९५ मिति:- २०४४/११/२९ जिल्ला:- सर्प गण्डा/पति:- आशमानन्पा/इता गज्जे गज्जे/प्रमा:- इता गज्जे			नाम:- श.पि.प्र:- न.प.नं:- मिति:- जिल्ला:- गण्डा/पति:- गज्जे/प्रमा:-			पञ्चारी		क	६३.३६	३	५	५	५६	२०३९/११/९		वाकिलदहुरोत १/कि नं:२९६/ले ५००-९-०-३		

गाउँ विकास समिति/नगरपालिका

जग्गादायको दस्तखत:

अभिनेको

दस्तखत:

पाना नं:- ११

नाथी निशानको

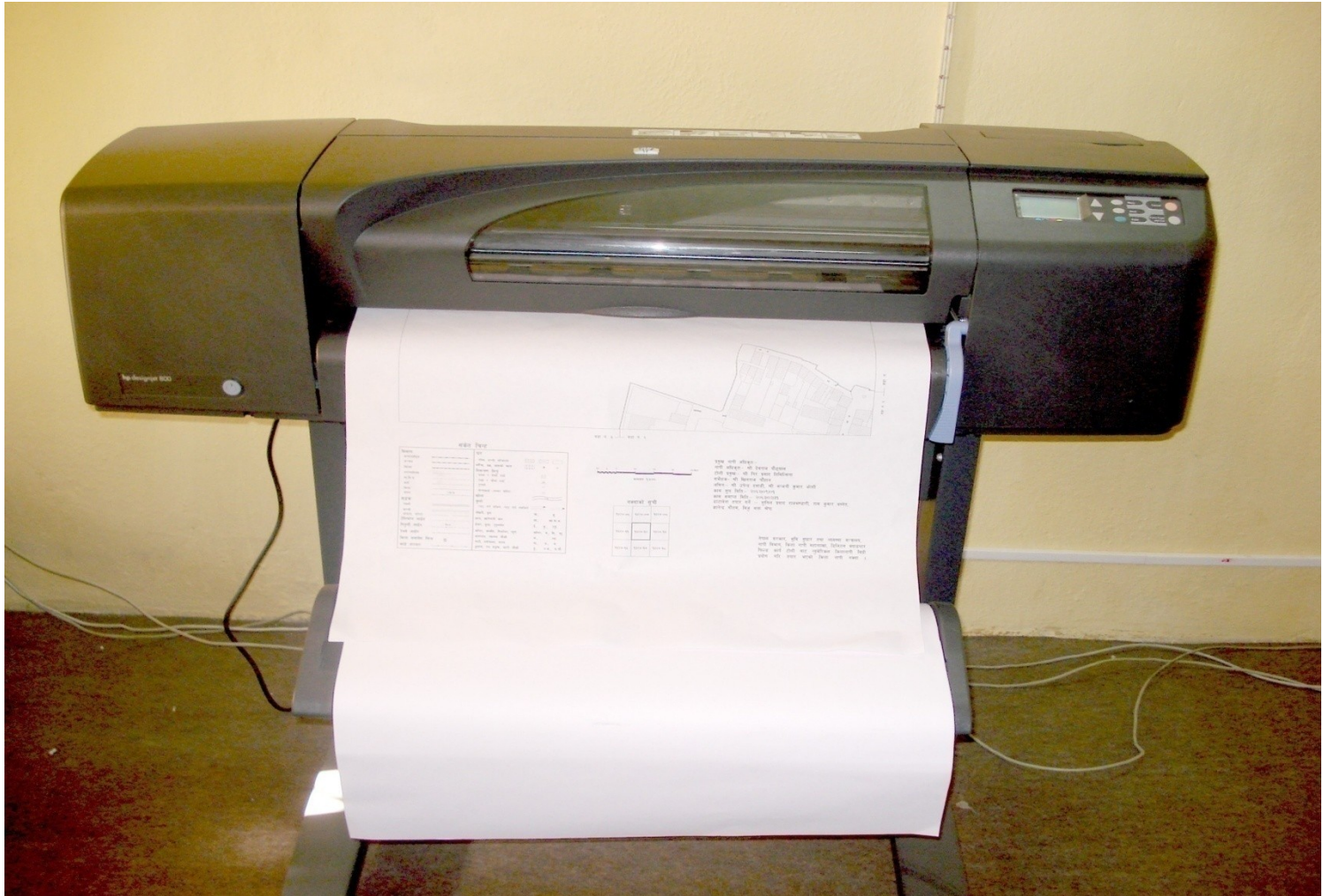
दस्तखत:

नाथी ठोली प्रमुखको

दस्तखत:

August 23, 2026 23 September, 2011

Map printing



August 23, 2026

Orthophoto Cadastre





Challenges in our context

- Initial Cost
- Expertise
- Lack of skilled manpower
- Lack of instruments
- Lack of hardware/software
- Legal obstruction
- Public awareness



Comparison between Graphical and Digital Cadastre

Graphical Cadastre

Advantage

Less chance of missing details

Easy to find mistake

Reduces office work

Economic

Better for mapping of land with low price

No need of expertise



Comparison between Graphical and Digital Cadastre

Disadvantage

Difficult to store documents

Less accurate

More field work

Tedious and time consuming

Difficulties for LIS

Difficult to change scale of map

Creates disputes

Comparison between Graphical and Digital Cadastre

Digital Cadastre

Advantage

- Accuracy
- Reliability
- Portability
- Analytical
- Seamless
- Scale less
- Fast
- Durable



Comparison between Graphical and Digital Cadastre Disadvantage

More initial cost

Needs skilled manpower

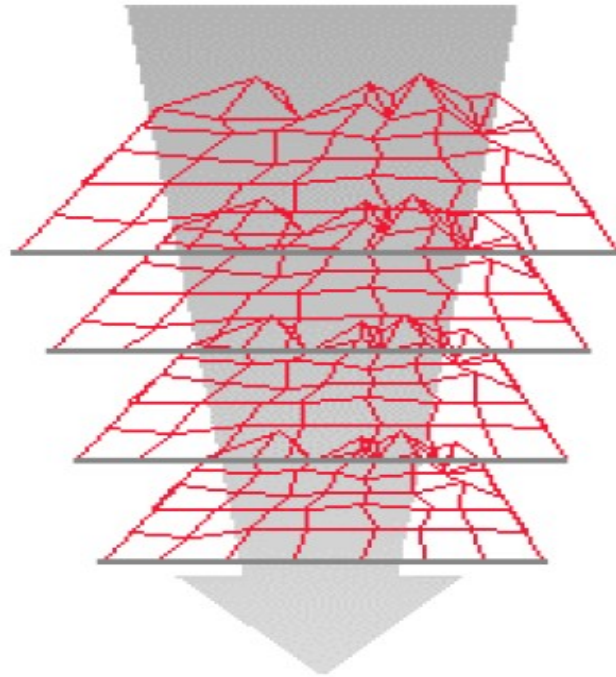
More office work

Needs field verification

Conclusion

- ✎ Now a days there is no alternate of Digital Cadastre
- ✎ Need of separate policy for conducting digital cadastral survey
- ✎ Need of incentive for increasing motivation of surveyor
- ✎ Public awareness (support) is very important
- ✎ Political commitment should be needed.

Modern approach to cadastral System



CADASTRE 2014

A VISION FOR A FUTURE CADASTRAL SYSTEM

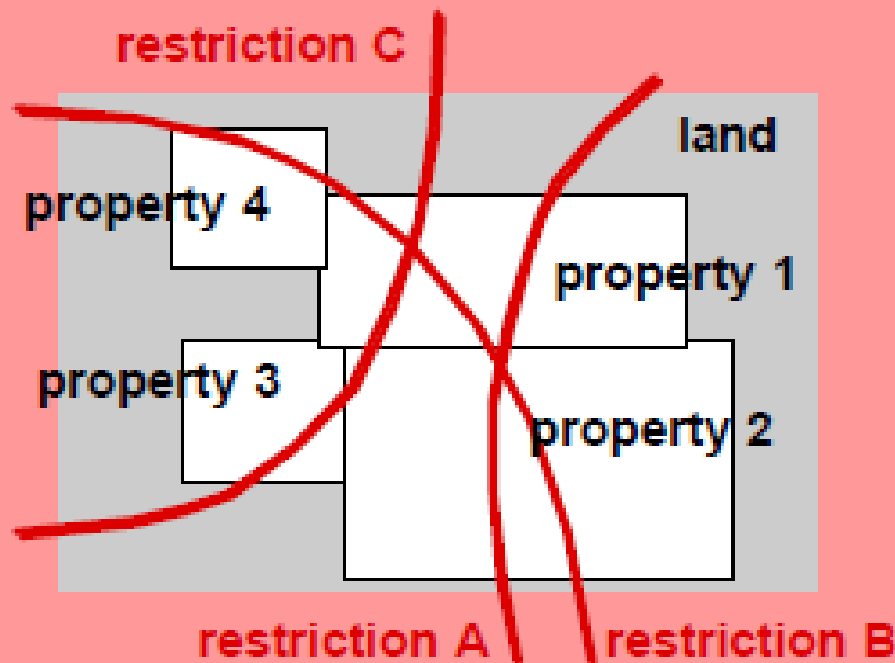


Modern approach to cadastral System

- Land parcels are considered as the key features rather than the land owner.
- Land Parcels are considered as the legal land object rather than the land property.
- Records are kept and prepared according as the parcels
- Cadastral modeling is emphasised
- There are six statements under vision of cadastral 2014 and seven principles to acheive the goal/objectives by 2014

Statement 1 on Cadastre 2014

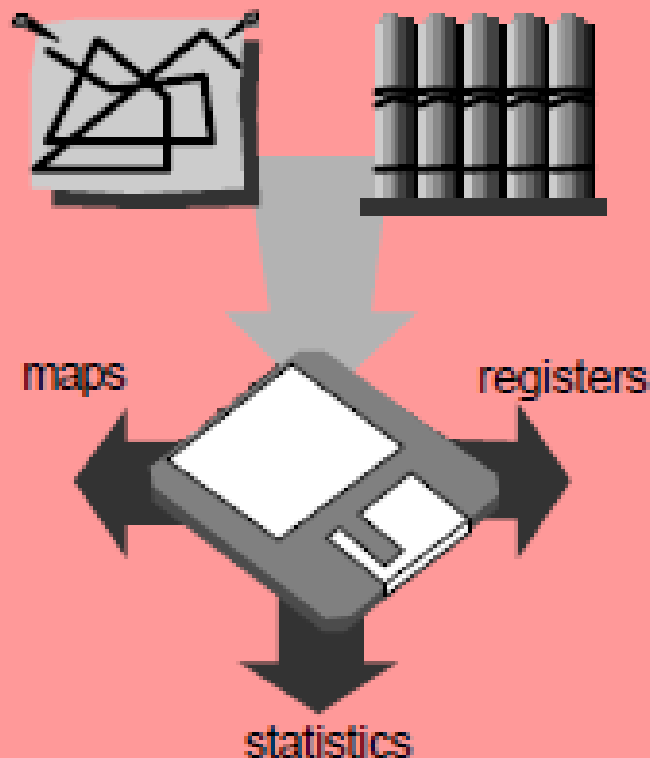
Cadastre 2014 will show the complete legal situation of land, including public rights and restrictions!



Comment: The population of the world is growing. The consumption of land is increasing. The absolute control of the individual or of legal entities of land is increasingly being restricted by public interest. To provide security of the land tenure, all facts about land must be made obvious by the cadastral system of the future.

Consequences: A new thematic model is necessary. Surveyors must take into consideration public law.

Statement 2 on Cadastre 2014

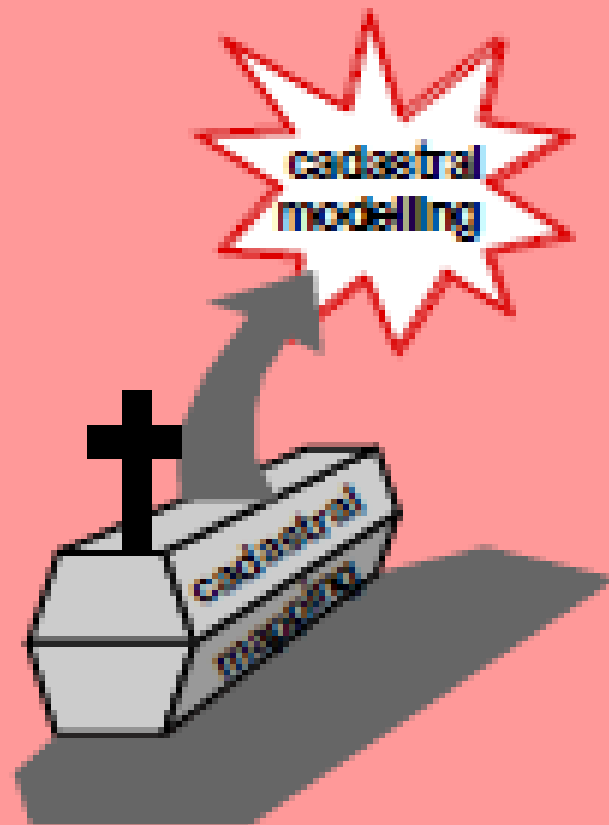


The separation between 'maps' and 'registers' will be abolished!

Comment: The separation was necessary because the available technology – paper and pencil – did not allow other solutions.

Consequences : The division of responsibilities between surveyor and solicitor in the domain of cadastre will be seriously changed.

Statement 3 on Cadastre 2014



The Cadastrol mapping will be dead!
Long live modelling!

Comment: Maps have always been models, but the available technology did not allow for the use of these models in a flexible manner. So in mapping flexibility had to be brought in by different scales. Different scales had to be represented by different data models.

Modern technology allows the creation of maps of different scales and registers in different forms from the the same data model.

Consequences: In 2014 there will be no draftmen and cartographers in the domain of cadastre.

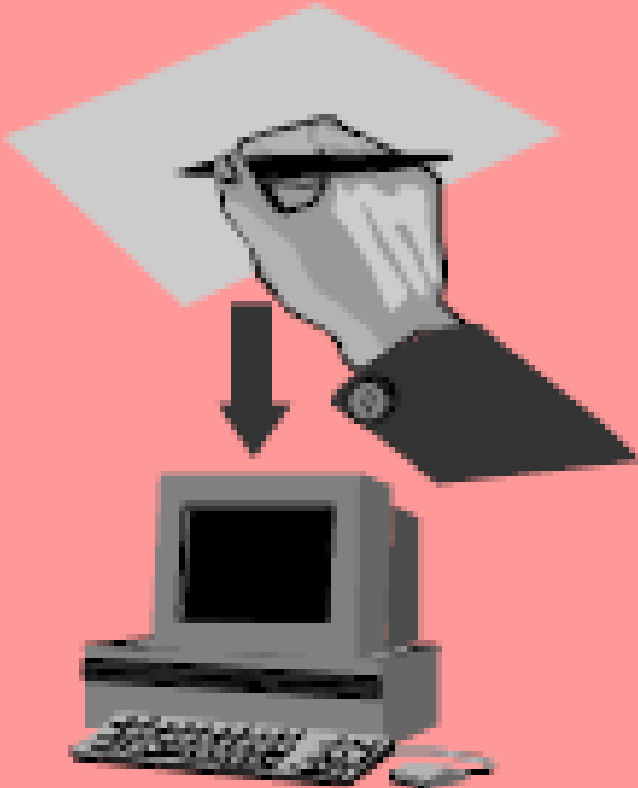
Statement 4 on Cadastre 2014

'Paper and pencil - cadastre' will have gone!

Comment: Geomatics technology will be the normal tool for cadastral work. Real low-cost solutions are only possible when this technology is used in combination with lean administrative procedures.

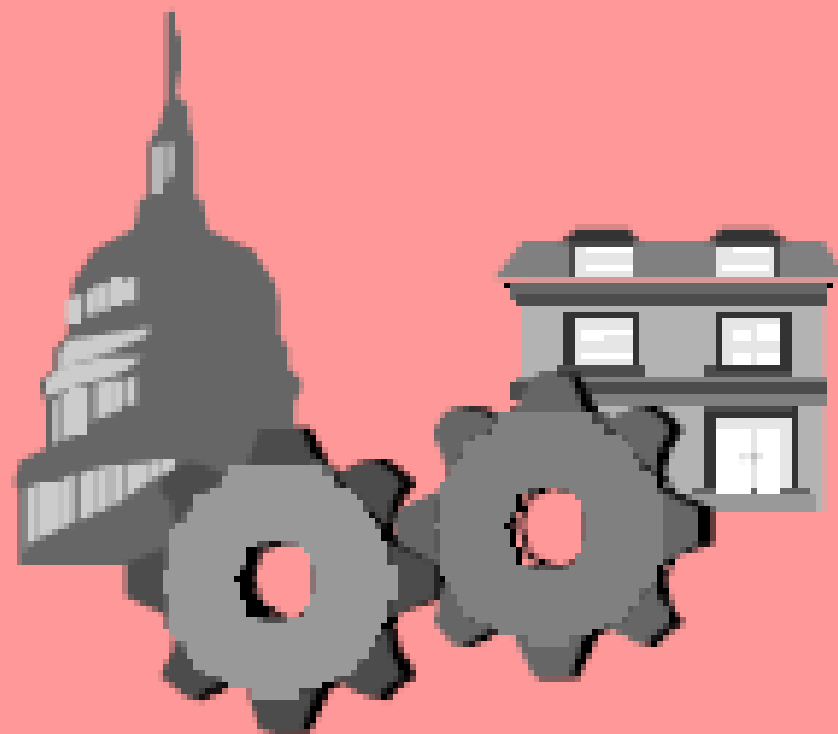
Developed, developing, and transitional countries need models of the existing situation to resolve the problems of population, environment and reasonable land use.

Consequences: The modern cadastre has to provide the basic data model. Surveyors all over the world must be able to think in models and to apply modern technology to handle such models.



Statement 5 on Cadastre 2014

Cadastre 2014 will be highly privatized! Public and private sector are working closely together!



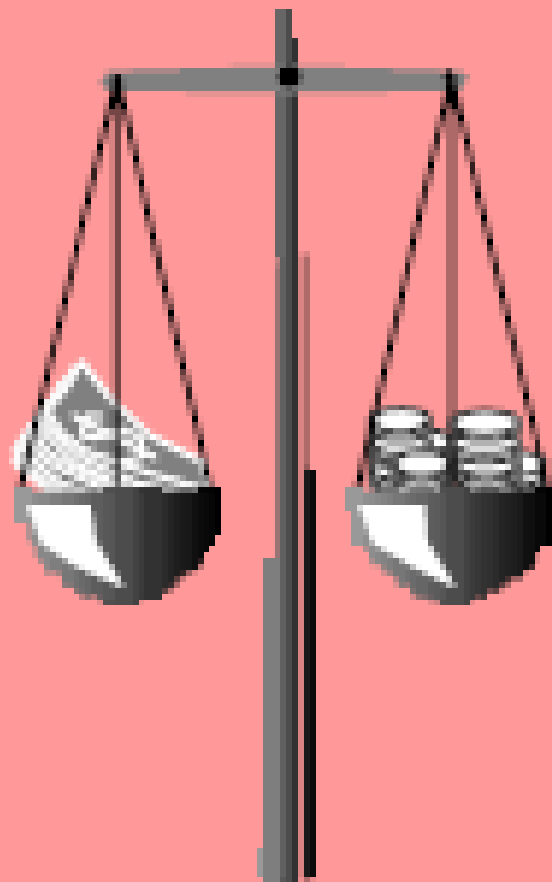
Comment: Public systems tend to be less flexible and customer oriented than those of private organizations.

Free economies demand flexibility in land markets, land planning and land utilization. Flexibility may be provided better by private institutions. For necessary security, however, public involvement is indispensable.

Consequences: The private sector will gain in importance. The public sector will concentrate on supervision and control.

Statement 6 on Cadastre 2014

Cadastre 2014 will be cost recovering!



Comment: Cadastral systems need considerable investment. But the land documented and secured by the cadastre represents a multiple of the investment. The investment and operation costs have to be paid back at least partially by those who profit.

Consequences: Cost/benefit analysis will be a very important aspect of cadastre reform and implementation. Surveyors will have to deal more with economic questions in future.

Parcel Editor Application

- ✎ Customization in ArcGIS 9x for cadastral application in digital environment.
- ✎ Full functionality of database for primary data collected by using Total Station (coordinate data)
- ✎ Preparation of different layers for multipurpose cadastral.
- ✎ Document preparation land ownership certificate, land ownership record, plot register, 7 day notification, field book etc.
- ✎ Parcel split/merge/print.

Spatial Application Extension (SAEx)

- **SAEx** is an application developed by **DoLIA** that could be Operated on ArcGIS 8.0
- Later it is upgraded to operate in upper versions also.
- It is developed specially for the Parcel Management of the cadastral maps and their included informations.
- The editing of the parcels are supported only for parcel merge and split.
- It adds specialized Tool Bar to the **ArcMap**

